Nested loops and third party tools using Python and QSGeo

In this exercise, you will combine your skills from the drone script with a Jupyter Notebook script used to cast shadows on a digital surface model (DSM) and a canopy digital surface model (CDSM).

The exercise consists of two parts:

- First task is to follow the tutorial found here: https://umep-docs.readthedocs.io/projects/tutorial/en/latest/Tutorials/PythonProcessing1.html . If you are using student computers, you do not need to follow the instructions on how to install and configure Jupyter Notebook as it is already there. Simply open a terminal in VSCode or open the OSGeo Shell and type **jupyter notebook**
- The second task is to cast shadows for a **10-day** period starting on your **birthdate** for each drone position point (**150 m radius**) and display the result (aggregated shadowmaps) in your notebook. Also, produce a map layout in QGIS where aggregated shadow maps are shown around each drone position. Add the map (image) at the end of your notebook. Here you use the London data (on Canvas, or from the previous exercise).

Hand in:

Hand in a QGIS map layout on Canvas including your 17 aggregated shadow maps.

Good luck!

TIPS:

- Make sure that you do not overwrite results from each loop iteration
- Your second task is solved by adding one additional (nested) loop by combining what you did earlier in the course